

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Industry Problem Task

Duration: 2.5hrs

1. Explain how Reinforcement Learning can be applied to optimize delivery routes in a logistics system. Identify the possible states, actions, rewards, and policy. Discuss how continuous learning improves efficiency over time. (5 Marks)

1. Understanding of Industry Problem (20%)

Logistics companies such as Amazon, DHL, and FedEx must deliver packages quickly while minimizing fuel costs and travel time. Traditional routing methods cannot effectively handle dynamic conditions like traffic congestion, road closures, weather changes, and urgent delivery requests. Reinforcement Learning (RL) provides an intelligent solution by enabling the system to continuously learn the best delivery routes from experience and adapt to changing environments.

2. Application of Engineering Concepts (25%)

The delivery route optimization problem is modeled as a Markov Decision Process (MDP).

- **State (S):** Vehicle location, traffic conditions, fuel level, weather, remaining delivery locations, and delivery deadlines.
- **Action (A):** Select the next route, reorder deliveries, avoid traffic, or choose an alternative road.
- **Reward (R):**

- +10 for successful on-time delivery.
- +5 for choosing the shortest route.
- -10 for delayed deliveries.
- -5 for high fuel consumption.

- **Policy (π):** A strategy that selects the best route to maximize long-term cumulative rewards.

Algorithms such as Q-Learning and Deep Q-Network (DQN) enable the agent to learn optimal routing decisions.

3. Solution Design (25%)

The proposed system consists of the following steps:

1. Collect real-time GPS, traffic, and weather data.
2. Represent the delivery environment as MDP states.
3. Train an RL agent using historical delivery data.
4. The trained agent selects the optimal route based on current conditions.
5. Delivery results are stored and used to improve future routing decisions through continuous learning.

This design minimizes delivery time, reduces operational costs, and increases customer satisfaction.

4. Use of Tools (15%)

Modern tools used include:

- Python
- TensorFlow/PyTorch
- Google Maps API
- GPS Tracking System
- OpenAI Gym
- Traffic and IoT Sensor Data

These tools provide real-time information required for effective RL training.

5. Analysis (10%)

Continuous learning enables the RL agent to adapt to changing traffic conditions, weather patterns, and customer demand. Compared to traditional shortest-path algorithms, RL reduces fuel consumption, improves delivery efficiency, and provides better long-term performance because it learns from every completed delivery.

6. Professionalism (5%)

Reinforcement Learning provides a practical and scalable solution for modern logistics systems. By integrating real-time data, intelligent routing, and continuous learning, organizations can reduce operational costs, improve delivery performance, and enhance customer satisfaction. The solution is reliable, industry-ready, and suitable for real-world logistics applications.

2. Apply Reinforcement Learning concepts to a fraud detection system in fintech. Define the state space, action space, and reward mechanism. Explain how exploration strategies improve detection accuracy.
(5 Marks)

1. Understanding of Industry Problem (20%)

Financial institutions process millions of online transactions every day, making fraud detection a critical challenge. Traditional rule-based systems often fail to detect new and evolving fraud techniques because they rely on predefined rules. Reinforcement Learning (RL) provides an adaptive solution by learning from transaction outcomes and continuously improving its fraud detection strategy. This helps reduce financial losses while minimizing false alarms for genuine customers.

2. Application of Engineering Concepts (25%)

The fraud detection problem is modeled as a Markov Decision Process (MDP).

- **State (S):** Transaction amount, customer purchase history, device ID, IP address, transaction location, merchant type, transaction frequency, and account behavior.
- **Action (A):**
 - Approve the transaction.
 - Reject the transaction.
 - Request additional verification (OTP or biometric authentication).
- **Reward (R):**
 - +20 for correctly identifying fraudulent transactions.
 - +10 for approving legitimate transactions.
 - -30 for approving fraudulent transactions.
 - -15 for rejecting genuine transactions (false positives).
- **Policy (π):** The RL agent learns the optimal decision policy that maximizes long-term rewards by accurately distinguishing fraudulent and genuine transactions.

To improve learning, exploration strategies such as ϵ -Greedy are used, where the agent occasionally explores alternative actions to discover new fraud patterns while exploiting previously successful detection strategies.

3. Solution Design (25%)

The proposed RL-based fraud detection system works as follows:

1. Collect real-time transaction and customer behavior data.
2. Represent each transaction as the current state.

3. The RL agent analyzes the transaction and chooses to approve, reject, or request verification.
4. The system receives feedback based on whether the decision was correct.
5. The reward is used to update the RL model, enabling continuous improvement in fraud detection accuracy.

This adaptive design allows the system to identify emerging fraud techniques without requiring frequent manual rule updates.

4. Use of Tools (15%)

The following modern tools and technologies can be used:

- Python
- TensorFlow or PyTorch
- Scikit-learn
- Banking Transaction Databases
- Apache Kafka (real-time transaction streaming)
- Cloud Platforms (AWS/Azure/GCP)

These tools support real-time processing, model training, and deployment in large-scale financial systems.

5. Analysis (10%)

Reinforcement Learning continuously improves fraud detection by learning from every transaction. Exploration strategies help the agent discover new fraud patterns, while exploitation uses previously successful decisions for accurate detection. Compared with traditional rule-based systems, RL offers higher adaptability, improved detection accuracy, reduced financial losses, and fewer false positives as fraud techniques evolve.

6. Professionalism (5%)

Reinforcement Learning provides an intelligent, scalable, and industry-ready solution for fraud detection in fintech. By combining real-time transaction analysis, continuous learning, and exploration strategies, the system enhances security, protects customer assets, reduces fraud-related losses, and builds trust in digital financial services.

3. Illustrate how Reinforcement Learning is used in a streaming platform for content recommendation. Discuss the impact of delayed rewards, user feedback, and changing preferences on learning performance. (5 Marks)

1. Understanding of Industry Problem (20%)

Streaming platforms such as Netflix, YouTube, and Spotify serve millions of users with diverse and constantly changing preferences. Recommending relevant content is challenging because user interests evolve over time, and poor recommendations can reduce user engagement and subscription retention. Reinforcement Learning (RL) provides an intelligent recommendation system that continuously learns from user interactions and improves personalized content suggestions.

2. Application of Engineering Concepts (25%)

The recommendation problem is modeled as a Markov Decision Process (MDP).

- **State (S):** User viewing history, watch time, search history, preferred genres, ratings, device type, and time of access.
- **Action (A):** Recommend a movie, TV show, song, or video to the user.
- **Reward (R):**
 - +10 if the user watches the recommended content completely.
 - +5 if the user likes or rates the content positively.
 - -5 if the user skips the recommendation.
 - -10 if the user leaves the platform after the recommendation.
- **Policy (π):** The RL agent learns a recommendation policy that maximizes long-term user engagement and satisfaction.

The system uses user feedback and delayed rewards to continuously improve recommendation quality.

3. Solution Design (25%)

The proposed RL-based recommendation system works as follows:

1. Collect user activity such as watch history, ratings, likes, and search behavior.
2. Represent the user's current preferences as the system state.
3. The RL agent recommends personalized content based on the learned policy.
4. User interactions (watch time, likes, skips, and ratings) are recorded as feedback.
5. The reward is used to update the RL model, enabling continuous improvement in future recommendations.

This adaptive design ensures that recommendations remain relevant as user preferences change over time.

4. Use of Tools (15%)

The following modern tools and technologies can be used:

- Python
- TensorFlow or PyTorch
- TensorFlow Recommenders
- Apache Spark (large-scale data processing)
- User Analytics Databases
- Cloud Platforms (AWS, Azure, GCP)

These tools support real-time recommendation generation and continuous model training.

5. Analysis (10%)

Reinforcement Learning improves recommendation performance by learning from delayed rewards and user feedback. Instead of relying only on immediate clicks, the system considers long-term user satisfaction based on watch time, ratings, and repeated platform usage. As user preferences change, the RL agent continuously updates its recommendation strategy, leading to higher engagement, increased watch time, and better customer retention compared to traditional recommendation methods.

6. Professionalism (5%)

Reinforcement Learning provides a practical and scalable solution for personalized content recommendation in streaming platforms. By combining delayed rewards, continuous user feedback, and adaptive learning, the system delivers highly relevant recommendations, improves user experience, increases subscription retention, and supports long-term business growth. This makes RL an effective and industry-ready solution for modern streaming services.

4. Evaluate the effectiveness of Reinforcement Learning in predictive maintenance compared to traditional rule-based approaches. Discuss risks and reliability concerns in industrial environments. (5 Marks)

1. Understanding of Industry Problem (20%)

Industries such as manufacturing, power plants, and oil & gas rely on machines that must operate continuously. Unexpected equipment failures can cause production delays, high repair costs, and safety risks. Traditional rule-based maintenance performs servicing at fixed intervals, which may lead to unnecessary maintenance or missed failures. Reinforcement Learning (RL) provides an intelligent solution by predicting the best time for maintenance based on machine conditions and continuously improving its decisions from operational data.

2. Application of Engineering Concepts (25%)

The predictive maintenance problem is modeled as a Markov Decision Process (MDP).

- **State (S):** Machine temperature, vibration level, pressure, operating hours, sensor readings, and equipment health status.
- **Action (A):**
 - Continue machine operation.
 - Schedule maintenance.
 - Replace faulty components.
 - Stop the machine for inspection.
- **Reward (R):**
 - +20 for preventing equipment failure.
 - +15 for reducing maintenance costs.
 - -30 for unexpected machine breakdowns.
 - -10 for unnecessary maintenance.

- **Policy (π):** The RL agent learns the optimal maintenance strategy that maximizes machine reliability while minimizing operational costs.

RL algorithms such as Q-Learning and Deep Q-Network (DQN) continuously improve maintenance decisions using real-time sensor data.

3. Solution Design (25%)

The proposed RL-based predictive maintenance system works as follows:

1. Collect real-time data from machine sensors such as temperature, vibration, and pressure.
2. Represent the machine's operating condition as the current state.
3. The RL agent decides whether to continue operation, inspect, or schedule maintenance.
4. The system receives rewards based on machine performance and maintenance outcomes.
5. The RL model continuously updates its policy, improving maintenance scheduling and reducing unexpected failures.

This solution minimizes downtime, extends equipment life, and lowers maintenance costs.

4. Use of Tools (15%)

The following modern tools and technologies can be used:

- Python
- TensorFlow or PyTorch
- IoT Sensors
- SCADA Systems
- Predictive Analytics Software
- Cloud Platforms (AWS, Azure, GCP)

These tools enable real-time monitoring, data collection, model training, and predictive maintenance.

5. Analysis (10%)

Compared to traditional rule-based maintenance, Reinforcement Learning continuously learns from equipment performance and adapts to changing operating conditions. It reduces unnecessary maintenance, prevents unexpected breakdowns, lowers operational costs, and improves equipment reliability. However, RL requires high-quality sensor data, sufficient training, and thorough validation before deployment in safety-critical industrial environments.

6. Professionalism (5%)

Reinforcement Learning offers a practical, intelligent, and industry-ready solution for predictive maintenance. By integrating real-time sensor data, continuous learning, and adaptive decision-making, the system improves equipment reliability, reduces downtime, enhances worker safety, and minimizes maintenance costs. This makes RL more effective than traditional rule-based maintenance systems in modern industries.

5. Analyze how Reinforcement Learning can be used to optimize transportation systems such as bus scheduling or route allocation. Define states, actions, and rewards, and discuss how real-time data improves performance. (5 Marks)

1. Understanding of Industry Problem (20%)

Public transportation systems often face problems such as traffic congestion, bus delays, overcrowding, and inefficient route planning. Fixed schedules cannot adapt to real-time changes in passenger demand or road conditions. Reinforcement Learning (RL) provides an intelligent solution by continuously learning from traffic and passenger data to optimize bus schedules and route allocation, improving service quality and reducing operational costs.

2. Application of Engineering Concepts (25%)

The transportation optimization problem is modeled as a Markov Decision Process (MDP).

- **State (S):** Bus location, passenger demand, traffic conditions, bus occupancy, route status, and time of day.
- **Action (A):**
 - Dispatch a bus.
 - Change the bus route.
 - Adjust departure or arrival schedules.
 - Allocate additional buses to crowded routes.
- **Reward (R):**
 - +20 for reducing passenger waiting time.
 - +15 for maintaining on-time bus schedules.
 - +10 for improving bus occupancy.
 - -20 for delays and traffic congestion.
 - -10 for unnecessary fuel consumption.
- **Policy (π):** The RL agent learns the optimal scheduling and routing policy that maximizes passenger satisfaction while minimizing travel time and operational costs.

Algorithms such as Q-Learning and Deep Q-Network (DQN) enable the system to make intelligent real-time decisions.

3. Solution Design (25%)

The proposed RL-based transportation management system works as follows:

1. Collect real-time data from GPS devices, traffic sensors, and passenger counting systems.
2. Represent the current transportation conditions as system states.
3. The RL agent selects the best route, schedule, or bus allocation based on the learned policy.

4. Passenger waiting time, travel time, and fuel usage are evaluated to generate rewards.
5. The RL model continuously updates its policy using real-time data, improving scheduling and routing decisions over time.

This adaptive solution reduces delays, improves service reliability, and optimizes resource utilization.

4. Use of Tools (15%)

The following modern tools and technologies can be used:

- Python
- TensorFlow or PyTorch
- Google Maps API
- GPS Tracking Systems
- IoT Traffic Sensors
- Cloud Platforms (AWS, Azure, GCP)

These tools support real-time data collection, route optimization, and model training.

5. Analysis (10%)

Reinforcement Learning continuously adapts to changing traffic conditions and passenger demand using real-time data. Compared to fixed scheduling methods, RL reduces waiting time, improves route efficiency, lowers fuel consumption, and increases passenger satisfaction. Continuous learning allows the transportation system to become more accurate and efficient over time.

6. Professionalism (5%)

Reinforcement Learning provides a practical, scalable, and industry-ready solution for intelligent transportation systems. By integrating real-time GPS data, traffic information, and adaptive learning, the system optimizes bus scheduling and route allocation, reduces operational costs, improves public transport reliability, and enhances the overall passenger experience.

Code of Conduct:

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sahith Sai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation